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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CC3001-1- CYBER SECURITY & FORENSICS
(For CCE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which block size is commonly used in AES?
A) 32 bits
B) 64 bits
C) 128 bits
D) 512 bits
2. Fast – flux DNS is often used in
A) Secure Communication
B) Firewall updates
C) Certificate issuance
D) Botnets and Malware
3. Which algorithm is commonly used in digital signatures?
A) RSA
B) DES
C) AES
D) RC4
4. The main purpose of DNS is:
A) Encrypt data
B) Map domain names to IP addresses
C) Provide firewall services
D) Generate digital signatures
5. The primary function of a firewall is:
A) Encrypt data
B) Control network traffic
C) Manage keys
D) Provide DNS services
6. An application gateway firewall is also known as:
A) Packet filter
B) IDS
C) Stateful filter
D) Proxy firewall
7. Which of the following is NOT an antifoensics technique?
A) Data wiping
B) Log manipulation
C) Load balancing
D) Encrypting files
8. Which header in HTTP may reveal proxy use?
A) X-Forwarded-For
B) Host
C) User-Agent
D) Content-Type
9. Rogue antivirus attackers profit mainly by:
A) Selling ads
B) Charging for fake licenses
C) Creating strong encryption
D) Repairing real infections
10. A key motivation for click fraud is:
A) Increasing revenue fraudulently
B) Encrypting traffic
C) Protecting DNS servers
D) Securing browsers
11. Which tool is often used for automated click fraud?
A) Firewalls
B) IDS
C) Botnets
D) Reverse proxy



CC3001-1

SEE – December 2025

12. Click fraud detection often relies on:
 - A) Encryption
 - B) Password hashes
 - C) VPNs
 - D) Traffic pattern analysis
13. Botnets often spread via:
 - A) Phishing emails
 - B) Firewalls
 - C) DNSSEC
 - D) HTTPS
14. A well-known botnet that used fast-flux was:
 - A) Zeus
 - B) Storm Worm
 - C) Conficker
 - D) Waledac
15. Which feature helps botnets stay resilient?
 - A) Redundant infected nodes
 - B) Single static server
 - C) Weak encryption
 - D) DNSSEC
16. Which of the following is NOT a use of botnets?
 - A) Distributed Denial-of-Service attacks
 - B) Click fraud
 - C) Data exfiltration
 - D) Antivirus signature updates
17. The first phase of the digital forensic life cycle is:
 - A) Analysis
 - B) Identification
 - C) Preservation
 - D) Reporting
18. The last step in the forensic life cycle is:
 - A) Documentation and reporting
 - B) Evidence collection
 - C) Seizure
 - D) Preservation
19. In the acquisition phase of the forensics life cycle, investigators:
 - A) Write final reports
 - B) Install antivirus software
 - C) Collect and preserve data
 - D) Encrypt logs
20. Network forensics focuses on:
 - A) Disk imaging
 - B) File carving
 - C) Registry cleaning
 - D) Analysis of network traffic

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain with an example how nonrepudiation ensures accountability in digital communication.	8	L*2	1	1,3
	b) Demonstrate with an example how encryption and decryption work using a simple key-based system.	8	L2	1	1,3
2.	a) Illustrate how the CIA triad (Confidentiality, Integrity, Availability) applies to securing online banking transactions.	8	L2	1	1,3
	b) Explain with an example how XOR can be used as a simple encryption technique.	8	L2	1	1,3
3.	a) Define antifoensics and list and explain three common goals attackers achieve using antifoensic techniques.	8	L2	2	2,3
	b) Explain how ICMP tunnelling can be used to bypass firewall restrictions and why it is difficult to detect.	8	L2	2	2,3
Unit – II					
4.	a) Explain how mobile malicious code differs from traditional computer malware.	8	L2	3	2,3
	b) Illustrate with an example how a user might unknowingly install mobile malicious code through a phishing message.	8	L2	3	2,3

CC3001-1

SEE – December 2025

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| 5. a) | Demonstrate how multi-factor authentication can reduce the success rate of phishing attacks on mobile devices. | 8 | L2 | 3 | 2,3 |
| b) | Describe the motivations behind click fraud and its implications for advertisers and publishers. | 8 | L2 | 3 | 2,3 |
| 6. a) | Describe what <i>token kidnapping</i> means in the context of privilege escalation attacks. | 8 | L2 | 4 | 1,3 |
| b) | Discuss how attackers exploit software vulnerabilities to gain administrator permissions. | 8 | L2 | 4 | 1,3 |

Unit – III

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|-------|--|---|----|---|-------|
| 7. a) | Explain the <i>need for computer forensics</i> in modern cybersecurity investigations. | 8 | L2 | 5 | 1,2,3 |
| b) | What is a <i>honeypot</i> ? List different types of honeypots used in cybersecurity. | 8 | L2 | 5 | 1,2,3 |
| 8. a) | Define <i>malicious code naming</i> and explain why standardization is necessary. | 8 | L2 | 5 | 1,2,3 |
| b) | Explain the <i>need for cyber laws in India</i> and how they address cyber threats and digital evidence admissibility. | 8 | L2 | 5 | 1,2,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
